Part 1: Python Debugging Exam (30%)

For each program, rewrite the incorrect lines on a separate sheet, indicating the original lines that have been replaced.

Exercise 1

```
age = input("Enter your age: ")

if age >= 18:
    print("You are an adult.")
```

Interpreter Error (after entering 20)

TypeError: '>=' not supported between instances of 'str' and 'int'

Exercise 2

```
num1 = 10
num2 = 0

try:
    result = num1 / num2
except:
    print("Division by zero error")
print(result)
```

Interpreter Error

NameError: name 'result' is not defined

Exercise 3

```
colors = ["red", "green", "blue"]
last_index=len(colors)
print("The last item is: ",colors[last_index])
```

Interpreter Error

IndexError: list index out of range

Part 2: Design and implement a class PIDController (70%)

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1. Your implementation must contain exactly two methods:

```
class PIDController:
    def __init__(self, kp, ki, kd):
        pass
    def update(self, error):
        pass
```

No additional public methods are allowed.

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2. The controller must implement the following discrete PID equation:

$$u(k) = K_p e(k) + K_i \sum_{i=0}^{k} e(i) + K_d(e(k) - e(k-1))$$

Where:

- $e(k)$ is the current error (provided as parameter to update)
- $e(k-1)$ is the previous error
- $\sum_{i=0}^{k} e(i)$ is the accumulated error (integral term)
- K_p, K_i, K_d are controller gains
- $u(k)$ is the control output returned by update()

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3. Example Usage

```
pid = PIDController(1.0, 0.1, 0.5)
u1 = pid.update(10)
u2 = pid.update(10)
```

The second call must:

- Use the stored previous error.
- Increase the accumulated integral term.
- Produce a different output than the first call.

Notes

- Assume fixed sampling time.
- No external libraries are required.
- Focus on correctness, clarity, and proper state handling.